Parametric Equations

Section 1: Plotting and Direction of Motion

1 For the parametric curve $x = 2t + 1, y = t^2 - 4$ for $-2 \le t \le 2$, find the initial point and terminal point.

a. Initial: $(-3, 0)$, Terminal: $(5, 0)$

b. Initial: $(1, -4)$, Terminal: $(5, 0)$

c. Initial: $(-3, 0)$, Terminal: $(1, -4)$

d. Initial: $(0, -3)$, Terminal: $(0, 5)$

2 How is orientation (direction of motion) shown on a parametric graph?

a. Arrows placed along the curve indicating the direction as the parameter t increases

b. By the slope of the curve

c. By the color of the line

d. By the quadrant of the initial point

3 Determine the direction of motion for $x = 3 - t, y = 2t$ as t increases.

a. Leftward and upward

b. Rightward and upward

c. Leftward and downward

d. Rightward and downward

Section 2: Eliminating the Parameter (Algebraic)

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4 Eliminate the parameter t from $x = t - 3, y = 2t + 1$.

a. $y = 2x + 7$

b. $y = 2x - 5$

c. $y = 2x - 7$

d. $y = x + 4$

5 Eliminate the parameter from $x = t^2, y = 3t$ for $t \geq 0$.

a. $x = \frac{y^2}{9}$ for $y \geq 0$

b. $y = 3\sqrt{x}$

c. Both of the above are correct

d. $x = 9y^2$

6 Eliminate the parameter from $x = e^t, y = 2e^{2t} - 1$.

a. $y = 2x^2 - 1$ for $x > 0$

b. $y = 2x - 1$

c. $y = e^{2x} - 1$

d. $y = 4x^2 - 1$

Section 3: Eliminating the Parameter (Trigonometric)

7 Eliminate the parameter from $x = 3\cos\theta, y = 3\sin\theta$.

a. $x^2 + y^2 = 9$

b. $x^2 + y^2 = 3$

c. $9x^2 + 9y^2 = 1$

d. $\frac{x^2}{9} - \frac{y^2}{9} = 1$

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8 Identify the Cartesian equation for $x = 4\cos\theta, y = 2\sin\theta$.

a. $\frac{x^2}{16} + \frac{y^2}{4} = 1$ (an ellipse)

b. $x^2 + y^2 = 20$

c. $\frac{x^2}{4} + \frac{y^2}{16} = 1$

d. $\frac{x^2}{16} - \frac{y^2}{4} = 1$

9 What curve is traced by $x = 3\sec\theta, y = 2\tan\theta$?

a. A hyperbola: $\frac{x^2}{9} - \frac{y^2}{4} = 1$

b. An ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

c. A parabola: $y = 2x^2$

d. A circle: $x^2 + y^2 = 13$

Section 4: Projectile Motion Applications

10 State the standard parametric equations for projectile motion launched with speed v_0 at angle θ from $(0, h_0)$ (with $g = 32$ ft/s^2).

a. $x(t) = (v_0\cos\theta)t, y(t) = -16t^2 + (v_0\sin\theta)t + h_0$

b. $x(t) = v_0 t, y(t) = -32t^2 + h_0$

c. $x(t) = (v_0\sin\theta)t, y(t) = -16t^2 + h_0$

d. $x(t) = (v_0\cos\theta)t, y(t) = (v_0\sin\theta)t$

11 A ball is kicked from ground level at 64 ft/s at an angle of 30°. At what time does it hit the ground?

a. 2 seconds

b. 1 second

c. 4 seconds

d. 3 seconds

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12 In the previous question, what is the horizontal range of the ball?

a. $64\sqrt{3}$ ft

b. 128 ft

c. $32\sqrt{3}$ ft

d. 64 ft

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